Explicit $SU(2)$ Heat-Kernel Initialization

Local derivative budgets, image-tail control, and a gauge-invariant exceptional event

Research dossier—one-plaquette theorem and block-interface audit

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Abstract

The preceding one-step effective-action theorem propagates five declared derivative budgets but does not initialize them for the exact nonlinear gauge action. This paper removes the existential constant from the radial one-plaquette part of that problem. With $SU(2)$ identified with the unit three-sphere and heat generator $\frac{1}{2}\Delta$, the character and Poisson-image representations are matched exactly. Pairing positive and negative images yields explicit uniform bounds on the first four derivatives of the image correction, including an analytic omitted-image tail. For $0 < t \le 0.2$ and class angle $\theta \le 1$, the resulting logarithmic-curvature certificate is

$$t \left\| \nabla^2 \left(-\log p_t - \frac{\theta^2}{2t} \right) \right\| \le 0.0824566, \qquad t \left\| D^3 \left(-\log p_t - \frac{\theta^2}{2t} \right) \right\| \le 0.149094.$$

The event $\theta(U_p) \ge \rho$ is conjugation invariant. Its one-plaquette probability is enclosed by an analytically integrated character series with a geometric truncation bound and explicit floating-roundoff allowance. A PROVED chain-rule theorem transports the radial constants through any curvature chart whose first three derivative norms are supplied. Establishing those norms on the full 2^4 quotient chart and converting plaquette tails into dependent block-polymer activities remain OPEN.

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1 Target, normalization, and scope

The one-step theorem of the preceding stage is exact but typed: it accepts numbers such as $\varepsilon_F, \varepsilon_X, \varepsilon_Y, \tau_F, \tau_X$ and returns a generated-action certificate. The earliest missing implication is to obtain such numbers from the nonlinear gauge action. This stage isolates the first factor in that implication:

$$\text{exact radial heat-kernel action} \longrightarrow \text{curvature-coordinate derivative budgets.}$$

The heat-kernel lattice action follows Menotti and Onofri [1]. Explicit heat-kernel eigenfunction and image representations on homogeneous spaces are part of the harmonic-analysis framework developed by Camporesi [2].

Identify $SU(2)$ with the unit sphere S^3. Every conjugacy class has a representative

$$U(\theta) = \operatorname{diag}(e^{i\theta}, e^{-i\theta}), \qquad 0 \le \theta \le \pi.$$

Haar probability has radial density

$$d\mu_{\mathrm{rad}}(\theta) = \frac{2}{\pi} \sin^2 \theta \, d\theta. \tag{1}$$

We fix heat generator $\frac{1}{2}\Delta_{S^3}$. The eigenvalue in the $(n+1)$-dimensional irreducible representation is $n(n+2)$, so the central heat kernel is

$$p_t(\theta) = \sum_{n=0}^{\infty} (n+1) e^{-n(n+2)t/2} \frac{\sin((n+1)\theta)}{\sin\theta}. \tag{2}$$

This convention fixes the principal local action as $\theta^2/(2t)$. Changing the metric or Laplacian convention changes every numerical time value and must be performed before comparison with another regulator.

The target is one plaquette and one logarithmic curvature chart strictly inside the cut locus. The result is C2: a rigorous restricted theorem. It is not a full block theorem, a large-field polymer theorem, or a mass-generation result. The Clay target remains the continuum construction and Hamiltonian gap described by Jaffe and Witten [4].

2 Poisson-image representation

Set $m = n+1$ in (2). Since $n(n+2) = m^2 - 1$,

$$p_t(\theta) = \frac{e^{t/2}}{\sin\theta} \sum_{m=1}^{\infty} m e^{-m^2 t/2} \sin(m\theta).$$

Poisson summation of the periodic Gaussian, followed by differentiation in θ, gives

$$p_t(\theta) = C(t) \frac{1}{\sin\theta} \sum_{k\in\mathbb{Z}} (\theta + 2\pi k) \exp\left[-\frac{(\theta+2\pi k)^2}{2t}\right], \qquad C(t) = \frac{e^{t/2}\sqrt{2\pi}}{2t^{3/2}}. \tag{3}$$

Factor the principal image:

$$\sum_{k\in\mathbb{Z}} (\theta + 2\pi k) e^{-(\theta+2\pi k)^2/(2t)} = \theta e^{-\theta^2/(2t)} (1 + q_t(\theta)). \tag{4}$$

Pairing k with $-k$, and writing $a_k = 2\pi k$, yields the analytic even function

$$q_t(\theta) = 2\sum_{k=1}^{\infty} e^{-a_k^2/(2t)} \left[\cosh\left(\frac{a_k\theta}{t}\right) - \frac{a_k}{\theta}\sinh\left(\frac{a_k\theta}{t}\right)\right]. \tag{5}$$

The apparent singularity at $\theta = 0$ is removable. Indeed,

$$\frac{a_k}{\theta}\sinh\left(\frac{a_k\theta}{t}\right) = \frac{a_k^2}{t}\int_0^1 \cosh\left(\frac{sa_k\theta}{t}\right) ds. \tag{6}$$

3 Explicit image derivative bounds

Fix $0 < \rho < \pi/2$ and $0 < t \le t_0$. For $j = 0, 1, 2, 3$, define

$$Q_j(t_0, \rho) := 2\sum_{k=1}^{\infty} e^{-a_k^2/(2t_0)+a_k\rho/t_0} \left(\frac{a_k}{t_0}\right)^j \left(1 + \frac{a_k^2}{t_0(j+1)}\right). \tag{7}$$

Lemma 3.1 (Uniform image bound, PROVED)**.** *Let*

$$t_0 \le \frac{(2\pi)^2 - 4\pi\rho}{8}. \tag{8}$$

Then

$$\sup_{0<t\le t_0}\sup_{0\le\theta\le\rho} |q_t^{(j)}(\theta)| \le Q_j(t_0, \rho), \qquad j = 0, 1, 2, 3.$$

After retaining images $1 \le k \le K$, the omitted series is bounded by the first omitted term divided by $1 - r_{j,K}$, where

$$r_{j,K} = \left(\frac{K+2}{K+1}\right)^{j+2} \exp\left[-\frac{(2\pi)^2(2K+3)}{2t_0} + \frac{2\pi\rho}{t_0}\right] < 1. \tag{9}$$

Proof. Differentiate (5) and use (6). The jth derivative introduces $(a_k/t)^j$ and s^j under the integral. Bounding hyperbolic sine and cosine by $e^{a_k\rho/t}$ and integrating s^j gives the kth summand of (7). Each summand has the form $t^{-p}e^{-c_k/t}$ with $p \le j + 1 \le 4$. Condition (8), using the first image, makes it increasing on $(0, t_0]$, so $t = t_0$ is the uniform endpoint. The polynomial ratio between consecutive k terms is at most $((k+1)/k)^{j+2}$; the Gaussian ratio gives (9), which decreases with k. Geometric summation proves the tail bound. □

Write

$$s_t(\theta) := -\log p_t(\theta) = \frac{\theta^2}{2t} + a(\theta) + b_t(\theta) + c(t), \tag{10}$$

where

$$a(\theta) = \log\frac{\sin\theta}{\theta}, \qquad b_t(\theta) = -\log(1 + q_t(\theta)).$$

If $Q_0 < 1$, direct differentiation gives

$$B_2 := \sup|b_t''| \le \frac{Q_2}{1-Q_0} + \frac{Q_1^2}{(1-Q_0)^2}, \tag{11}$$

$$B_3 := \sup|b_t'''| \le \frac{Q_3}{1-Q_0} + \frac{3Q_1Q_2}{(1-Q_0)^2} + \frac{2Q_1^3}{(1-Q_0)^3}. \tag{12}$$

The geometric term has

$$A_2(\rho) := \sup_{[0,\rho]} |a''| = \csc^2 \rho - \rho^{-2}, \qquad A_3(\rho) := \sup_{[0,\rho]} |a'''| = 2\rho^{-3} - \frac{2\cos\rho}{\sin^3\rho}. \tag{13}$$

The equalities follow from the positive-coefficient expansions of $\csc^2\theta - \theta^{-2}$ and its derivative on this interval.

4 Local curvature-action certificate

Let $X \in \mathfrak{su}(2) \cong \mathbb{R}^3$ and $\theta = \|X\|$. A radial function $u(\theta)$ has Hessian eigenvalues $u''(\theta)$ in the radial direction and $u'(\theta)/\theta$ in the two tangential directions.

Theorem 4.1 (Explicit local heat-kernel initializer, PROVED)**.** *Assume Lemma 3.1, $Q_0 < 1$, and $\|X\| \le \rho$. Then for all $0 < t \le t_0$,*

$$t\left\|\nabla_X^2 s_t(X) - \mathrm{I}/t\right\| \le \varepsilon_{\mathrm{HK}} := t_0(A_2 + B_2). \tag{14}$$

Moreover,

$$t\left\|\nabla_X\left(s_t(X) - \frac{\|X\|^2}{2t}\right)\right\| \le g_{\mathrm{HK}} := \rho\varepsilon_{\mathrm{HK}}, \tag{15}$$

and

$$t\left\|D_X^3\left(s_t(X) - \frac{\|X\|^2}{2t}\right)\right\| \le \tau_{\mathrm{HK}} := 4t_0(A_3 + B_3). \tag{16}$$

Proof. Apply (11) and (13) to the radial Hessian. Since the remainder derivative vanishes at zero, the mean-value theorem bounds the tangential eigenvalue and gives (15). For a radial C^3 function u, set

$$\beta = \frac{u''}{\theta} - \frac{u'}{\theta^2}.$$

If $M_3 = \sup|u'''|$, integration gives $|\beta| \le M_3/2$. The Euclidean third derivative tensor is the sum of a radial tensor with coefficient $u''' - 3\beta$ and three mixed tensors with coefficient β. Its operator norm is therefore at most $|u''' - 3\beta| + 3|\beta| \le 4M_3$, proving (16). □

For $(t_0, \rho) = (0.2, 1)$ the analytic quantities are

Quantity	Value
A_2	0.4122829274
A_3	0.1863671214
Q_0	2.3943×10^{-27}
Q_3	1.8840×10^{-23}
$\varepsilon_{\mathrm{HK}}$	0.08245658549
τ_{HK}	0.1490936971

The image contribution is much smaller than the group-Jacobian correction in this window; the implementation nevertheless retains its analytic tail.

5 Composition with a curvature chart

Let $X(A)$ be a declared plaquette-log map on a link-coordinate chart. Write $D_0 = DX(0)$ and suppose

$$\|D_0\| \le M_0, \quad \|DX\| \le M_1, \quad \|DX - D_0\| \le \delta_1, \quad \|D^2X\| \le M_2, \quad \|D^3X\| \le M_3. \tag{17}$$

Proposition 5.1 (Chart propagation, PROVED)**.** *On a region where $\|X(A)\| \le \rho$, the Hessian remainder of $s_t \circ X$ relative to $D_0^* D_0/t$ is bounded by*

$$\varepsilon_{\text{chart}} = \delta_1(M_1 + M_0) + \varepsilon_{\text{HK}} M_1^2 + (\rho + g_{\text{HK}})M_2. \tag{18}$$

Its scaled third derivative is bounded by

$$\tau_{\text{chart}} = \tau_{\text{HK}} M_1^3 + 3(1 + \varepsilon_{\text{HK}})M_1 M_2 + (\rho + g_{\text{HK}})M_3. \tag{19}$$

Proof. Use

$$D^2(s_t \circ X) = DX^*(D^2 s_t)DX + \langle Ds_t, D^2X\rangle$$

and $\|DX^*DX - D_0^*D_0\| \le \delta_1(M_1 + M_0)$ to obtain (18). The third-order chain rule is

$$D^3(s_t \circ X) = D^3 s_t[DX]^3 + 3D^2 s_t[D^2X, DX] + Ds_t[D^3X].$$

Theorem 4.1 gives $t\|D^2 s_t\| \le 1 + \varepsilon_{\text{HK}}$ and $t\|Ds_t\| \le \rho + g_{\text{HK}}$, proving (19). □

For the identity curvature chart, these formulas return exactly ε_{HK} and τ_{HK}. For a block, a declared aggregation operator norm must multiply the per-plaquette bounds. It cannot be silently replaced by either independence or raw plaquette count. A stress example with $(M_0, M_1, \delta_1, M_2, M_3) = (1, 1.1, 0.1, 0.2, 0.3)$ and aggregation factor 6 gives Hessian budget $3.16 > 2$. Those chart values are illustrative, not certified BCH constants; the failure demonstrates why their sharp derivation is the next proof obligation.

6 Gauge-invariant exceptional probability

Define the one-plaquette exceptional event

$$E_{\rho,p} = \{U_p : \theta(U_p) \ge \rho\}. \tag{20}$$

Since class angle is invariant under $U_p \mapsto gU_pg^{-1}$, this event is gauge invariant.

Combining (2) and (1) gives

$$\mathbb{P}_t(E_{\rho,p}) = \frac{2}{\pi}\sum_{n=0}^{\infty}(n+1)e^{-n(n+2)t/2} I_{n+1}(\rho), \tag{21}$$

where

$$I_1(\rho) = \frac{\pi - \rho}{2} + \frac{\sin 2\rho}{4}, \tag{22}$$

$$I_m(\rho) = -\frac{1}{2}\left(\frac{\sin((m-1)\rho)}{m-1} - \frac{\sin((m+1)\rho)}{m+1}\right), \qquad m \ne 1. \tag{23}$$

Proposition 6.1 (Character-tail enclosure, PROVED)**.** *After summing* $0 \le n \le N$*, the absolute omitted tail in* (21) *is at most*

$$R_N(t) = \frac{2(N+2)e^{-(N+1)(N+3)t/2}}{1 - r_N(t)}, \qquad r_N(t) = \frac{N+3}{N+2} e^{-(2N+5)t/2}, \tag{24}$$

provided $r_N(t) < 1$.

Proof. Equations (22)–(23) are the elementary integrals of $\sin(m\theta)\sin\theta$ over $[\rho, \pi]$. Their absolute values are at most π, so the omitted nth contribution is at most $2(n+1)e^{-n(n+2)t/2}$. The ratio of consecutive majorants decreases with n and equals $r_N(t)$ at the first omitted index. Sum the geometric majorant. □

At $\rho = 1$ and $N = 80$, the analytic truncation is far below the explicitly recorded floating-roundoff allowance:

t	$\mathbb{P}_t(\theta \ge 1)$	analytic series tail
0.20	0.1474007238	1.74×10^{-290}
0.10	0.01580367907	1.69×10^{-144}
0.05	0.0001437428033	1.70×10^{-71}

The code reports an interval rather than the midpoint alone. The probability is a genuine one-plaquette large-field datum. Neighboring plaquettes share links; therefore these numbers cannot be tensorized into a block or polymer activity without a dependence theorem.

7 Executable verification

The implementation performs fourteen positive and hostile checks:

- character and Poisson-image ratios agree to floating precision at three nonzero class angles;
- omitted image tails remain positive even below the ordinary floating range, with their logarithmic bounds retained;
- decreasing t_0 improves both local derivative budgets;
- the character probability interval contains a longer-series reference;
- probability is normalized at $\rho = 0$ and decreases with the calibrated radius;
- the identity chart preserves the radial budgets exactly;
- cut-locus radii, excessive time ranges, invalid chart norms, and nonpositive aggregation factors fail closed.

This is executable analytic bookkeeping rather than formal proof-assistant verification. Image and character truncation errors are bounded analytically; the partial sums use double precision with a separately recorded roundoff allowance.

8 Frontier audit and next theorem

ID	Required implication	Status
H1	Exact radial heat-kernel derivative initializer and one-plaquette exceptional probability	PROVED
F1	Explicit transported plaquette-log, Coulomb-slice, retained/fiber projection, and overlap constants on the full 2^4 block	OPEN
F2	Sharp block incidence aggregation preserving the Stage-06 fiber margin	OPEN
F3	Dependent gauge-invariant bad-block polymer estimate, including center flux, holonomy, exceptional orbits, boundaries, and topology	OPEN
F4	Initialization of all five Stage-06 components for generated localized interactions	OPEN
F5	Cutoff-uniform repeated RG closure and retained-base massive crossover	OPEN
F6	Nontrivial regulator-independent OS continuum theory and Hamiltonian mass gap	OPEN

The result changes the location of the first missing implication. The radial heat kernel itself no longer contributes an existential C_B: its geometric and image corrections are explicit. The main local uncertainty is now the representation map from 216 noncommuting plaquette holonomies, through a gauge slice and coarse retained data, into the $51 + 357$ quotient/fiber coordinates. The next bounded theorem should derive those block constants using the exact $SU(2)$ quaternion product and logarithm, rather than a formal BCH remainder. It should then insert the resulting numbers into the Stage-06 inequalities.

Even if that local block closes, the one-plaquette tail must still be replaced by a dependent polymer activity. Large fields, topology, repeated scales, infrared crossover, continuum existence, reflection positivity, observable density, and spectral transfer remain indispensable before any mass-gap claim.

References

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